97

Diagnostic and Interventional Nephrology

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A variety of procedures are essential to the care of nephrology patients and include ultrasound, kidney biopsy, insertion of hemodialysis (HD) and peritoneal dialysis (PD) catheters, creation of arteriovenous fistulas (AVF), and diagnostic and interventional procedures to maintain HD access function. These procedures have traditionally been performed by other specialists but are increasingly performed by nephrologists. The field of diagnostic and interventional nephrology is most developed in the United States, where the American Society of Diagnostic and Interventional Nephrology (ASDIN) (https://www.asdin.org) has established training standards and certification procedures. This chapter covers insertion of dialysis catheters and interventions in vascular access, focusing on the applications of these procedures and their performance by nephrologists. Kidney biopsy is covered in Chapter 7, and placement of AVF and arteriovenous grafts (AVG) in Chapter 96. The role and interpretation of ultrasound are further covered in Chapters 5 and 6.

PERITONEAL DIALYSIS CATHETERS

Successful PD depends on proper catheter insertion and management, which can be done safely and successfully by nephrologists,[1-3] thus facilitating uptake of PD. Tenckhoff catheters are the most popular chronic peritoneal access devices; they are constructed of silicone rubber with a 5-mm external diameter and internal diameters of 2.6 to 3.5 mm and numerous intraperitoneal side holes. The intraperitoneal portion can be straight, straight with perpendicular silicone disks, or curled. Several design modifications have been developed in attempts to diminish outflow obstruction, including a version sold in Europe with a weighted tip. The subcutaneous portion is either straight or bent and has one or two extraperitoneal Dacron cuffs that prevent fluid leaks and bacterial migration around the catheter. One type of catheter has a silicone ball and Dacron disc at the parietal peritoneal surface. The subcutaneous portion is curved to give a lateral or downward direction of the exit site. An upward-directed exit site collects debris and fluid, increasing the risk for exit site infection. The method of catheter placement has more effect on the outcome than does the catheter type.

Catheter Insertion

There are four techniques for PD catheter insertion: dissection (surgical), the Seldinger technique (blind or with fluoroscopy), peritoneoscopic, and laparoscopic.[4] The Seldinger technique with fluoroscopy is the procedure used by most nephrologists, because it uses the same fluoroscope and some disposables used in placement of central venous catheters and angioplasty of AVGs and AVFs. Peritoneoscopic insertion is a single-puncture technique using a small (2.2-mm diameter) optical peritoneoscope for direct inspection of the peritoneal cavity and identification of a suitable site for the optimal intraperitoneal portion of the catheter. Peritoneoscopic placement is performed using local anesthesia (sometimes with conscious sedation) and manual infusion of about 1 L of air to allow visualization of the anterior peritoneal surfaces. This technique is not used by many nephrologists because it involves specialized training and equipment including the peritoneoscope, a video camera to display images, and a table allowing Trendelenburg positioning of the patient.

Laparoscopic techniques are performed with the patient under general anesthesia using larger scopes, multiple insertion sites, and automated gas infusion (usually CO_2). Both peritoneoscopic and laparoscopic techniques allow direct visualization of anterior peritoneal surfaces, to identify adhesions and hernias, and to choose the optimal location of the PD catheter. Laparoscopic placement allows advanced techniques for improving catheter function including omentopexy, correction of hernias, and removal of adhesions, which improve the long-term success of PD catheters.[5,6]

Any of the methods for PD catheter placement can be successful if the physician performing placement has sufficient experience with the technique. Randomized and nonrandomized studies have documented that the peritoneoscopic and fluoroscopic Seldinger techniques can result in fewer catheter complications (infection, outflow failure, pericatheter leak) and improved catheter survival compared with surgical placement.[2,7] The superior results with peritoneoscopic placement may relate to direct visualization of the abdominal cavity, less tissue dissection, and avoidance of general anesthesia. Because tissue dissection is minimal during placement, the catheter can be used for intermittent dialysis immediately, although there is an increased risk of peri-catheter leaks if the catheter is used immediately and continually for fluid exchange.[8,9]

For peritoneoscopic insertion (Fig. 97.1), a small skin incision (2–3 cm) is made and dissection of subcutaneous tissue down to the rectus muscle.[10] The anterior rectus sheath is identified but not incised. A preassembled cannula with trocar and a spiral sheath is then inserted at a 45-degree angle into the abdominal cavity through the rectus muscle toward the pelvis (see Figs. 97.1A and 97.2). The trocar is then removed and replaced by the 2.2-mm diameter peritoneoscope to confirm the intraabdominal position of the cannula (see Fig. 97.1B). Air is then infused (600–1000 mL) to separate the visceral peritoneum and parietal peritoneum, using a sterile syringe or rubber bulb. Alternatively, a Veress needle can be used for first insertion to the peritoneum, but this needle does not allow inspection of the peritoneal surfaces before gas infusion.[11] During peritoneoscopy, bowel loops, mesentery, the dome of the bladder, and intraabdominal adhesions are identified. The cannula and surrounding spiral sheath are advanced toward the pelvis, avoiding any areas of peritoneal adhesions (see Fig. 97.1C). The cannula and the peritoneoscope are then removed, the spiral sheath is dilated to 6-mm diameter (see Fig. 97.1D), and the catheter is inserted through the sheath while stiffened by a stylet (see Fig. 97.1E).

Steps for the Insertion of a Peritoneal Dialysis Catheter by Peritoneoscopy

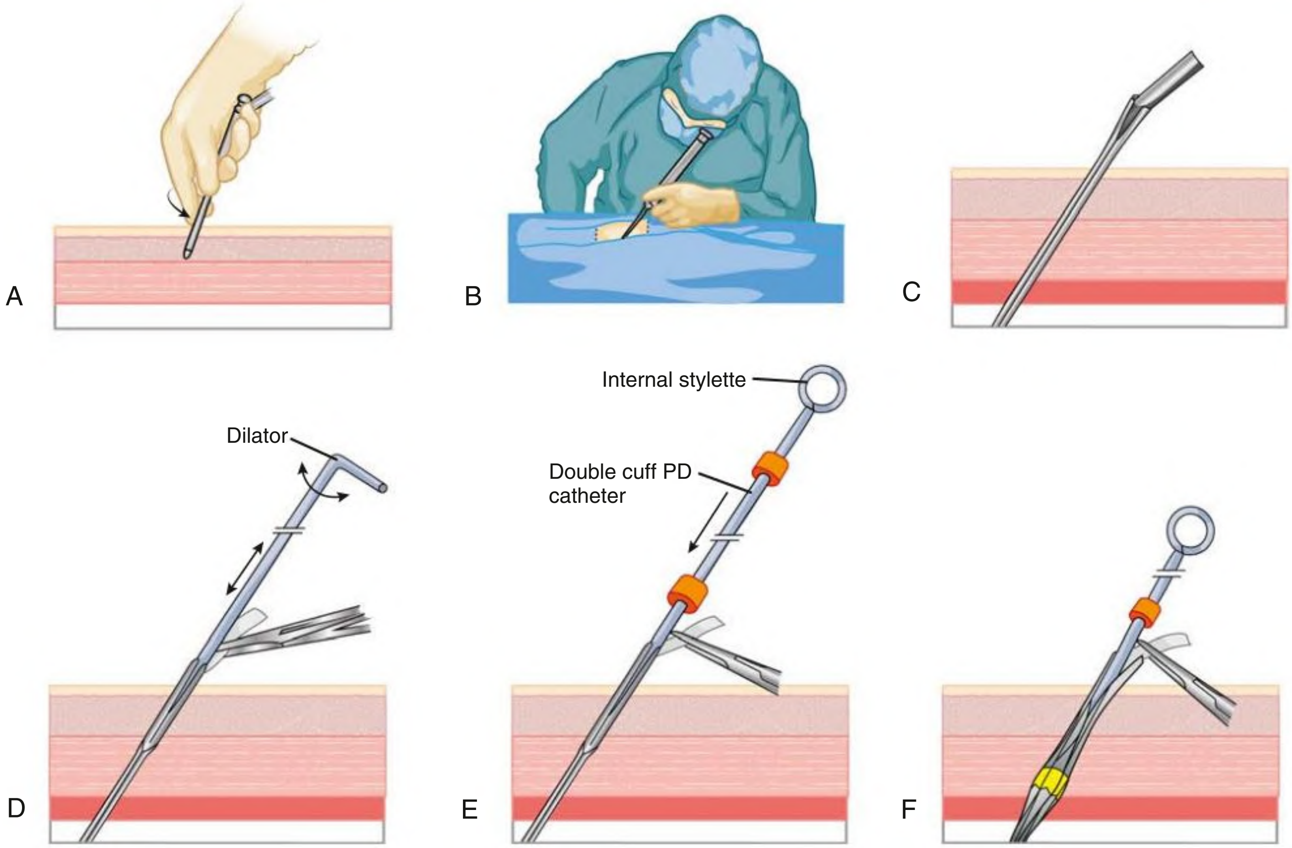

Fig. 97.1 Steps for the Insertion of a Peritoneal Dialysis (PD) Catheter by Peritoneoscopy. (A) A trocar and cannula with a sheath are inserted into the abdominal cavity. (B) A peritoneoscope is passed through and locked into the cannula. (C) The sheath has been passed into the abdominal cavity and the peritoneoscope and cannula removed sequentially. (D) The sheath is secured with forceps while it is being dilated. (E) A PD catheter (with double cuff) is passed through the dilated sheath by use of an internal stylet. (F) The deep cuff is implanted into the rectus muscle. (Modified from Y-Tec Instructions. *Laparoscopic and Peritoneoscopic Placement of Peritoneal Dialysis Catheters*. Medigroup Inc. [division of Janin Group, Inc.]; 2004:1–5.)

The deep cuff is implanted into the rectus muscle using an implanter tool without dissection of the anterior rectus sheath or the muscle (see Fig. 97.1F). A tunnel and an exit site are created, and the superficial cuff is implanted into the subcutaneous tissue. The dermis is sutured with absorbable material, and the epidermis is closed with nylon. No sutures are placed on the external rectus sheath or at the skin exit site. Instead of using a trocar, the cannula can be advanced into the peritoneal space over a 0.035-inch guidewire by placing a 7-mm dilator inside the cannula. This step is useful to inspect the peritoneum during a fluoroscopic placement, in case the guidewire does not progress as expected within the peritoneum.

The Seldinger technique using fluoroscopy begins with blunt dissection down to the level of the lateral border of the rectus sheath. An 18-gauge needle with internal blunt stylet or a 22-gauge needle from a 5-French micropuncture set is inserted at an angle of 45 degrees, directed toward the lower pelvis into the peritoneum (ultrasound is helpful to identify thickness of the rectus muscle and presence of adhesions to the parietal peritoneum).[12] The location of the needle within the peritoneal cavity is confirmed by injecting 1 to 5 mL of contrast material, which forms a classic "spider web" appearance on fluoroscopy as dye moves into small spaces between bowel loops and

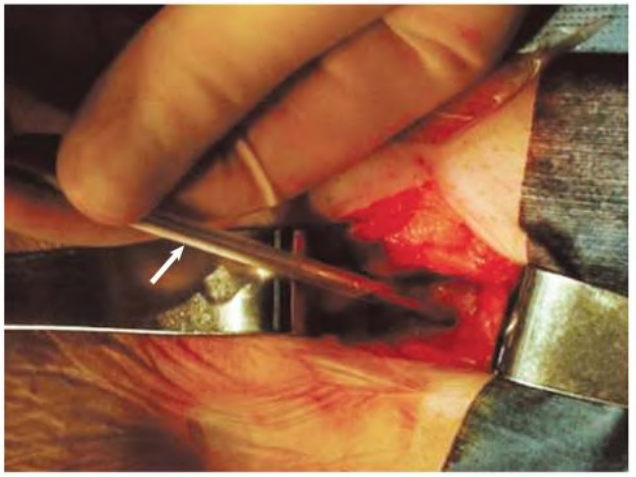

Fig. 97.2 Peritoneoscopic Insertion of a Peritoneal Dialysis Catheter. During peritoneoscopic insertion of a peritoneal dialysis catheter, a Quill guide trocar and cannula *(arrow)*, with its wrapped spiral sheath, is being inserted through the rectus muscle under local anesthesia.

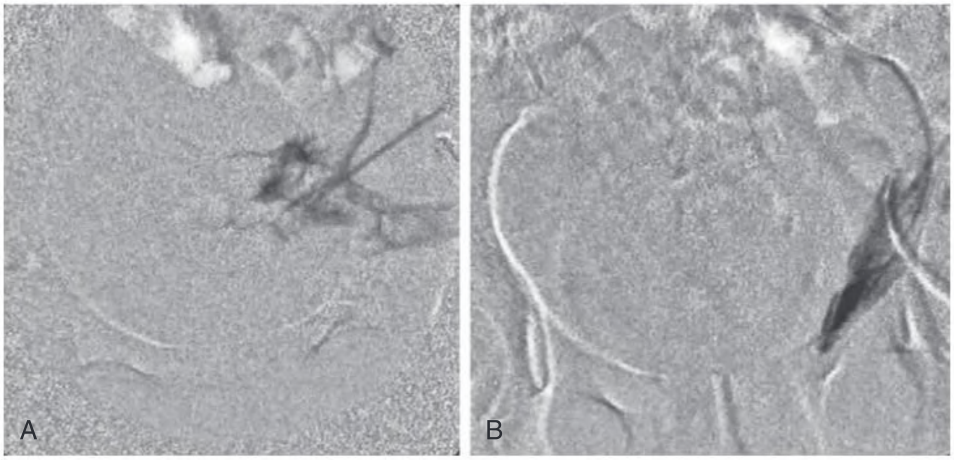

Fig. 97.3 (A) Fluoroscopic image after injecting a few milliliters of radiopaque dye through an 18-gauge needle after insertion through the rectus muscle and into the peritoneal space. Dye moves quickly away from the needle, and the image is not static. (B) Image after injecting a few milliliters of dye into an 18-gauge needle after insertion through the rectus muscle, with the tip of the needle in the preperitoneal space. Dye collects around the tip of the needle, at least one flat surface outlines the parietal peritoneum, and the image changes very little with time. (From Ash S, Sequeira A, Narayan R. Imaging and peritoneal dialysis catheters. *Semin Dial.* 2017;30[4]:338–346.)

parietal peritoneum (Fig. 97.3A).[12] If the tip of the needle is in the preperitoneal space, the dye collects around the tip of the needle, forms an outline of the parietal peritoneum, and appears static over a minute or so (Fig. 97.3B). If the tip of the needle appears to be preperitoneal, advancing the needle under fluoroscopic visualization allows the tip to penetrate the parietal peritoneal surface as confirmed by injection of a few milliliters of dye. If a micropuncture needle is used, an 0.018-inch guidewire is then inserted through the needle under fluoroscopy. Once the wire is in the lower pelvis, a 5-French catheter and internal dilator are advanced over the wire. Contrast material is again injected through the catheter to confirm the position. If an 18-gauge needle is used, a 0.035-inch guidewire is passed directly through the needle. Dilators are advanced sequentially over the guidewire up to the final 18-French dilator and peel-away sheath. The guidewire is removed, and the PD catheter with an internal metal stylet is advanced through the sheath, splitting the sheath as the deep cuff advances. This cuff is pushed into the rectus muscle while the sheath is in place, and the sheath is then removed around the cuff and catheter. The catheter is tunneled laterally with a tunneling tool (Fig. 97.4).[13,14] (For additional details, see reference[15].)

Fig. 97.4 **Subcutaneous Tunnel Being Created for a Peritoneal Dialysis Catheter.** With use of a disposable tool, a subcutaneous tunnel is created for the catheter. The superficial cuff (shown) will be implanted in the subcutaneous tunnel.

Burying (Embedding) the Peritoneal Dialysis Catheter

If the catheter will not be used immediately, the external portion can be buried under the skin for weeks to months before it is exteriorized and used. The catheter is placed in the usual manner, tunneled through a 1-cm skin exit site, filled with heparin solution (1000 U/mL), and blocked with a plug. The outside tip is then directed back into the exit site and tunneled through the subcutaneous space in a direction toward midline, caudal to the umbilicus. In the original publications, the catheter was tied off with silk suture and coiled into a pouch under the exit site.[16] Direct line tunneling of the external portion of the catheter (described earlier) avoids creation of a pocket just below the exit site and creates a normal exit site anatomy when exteriorized.

The primary incision and exit site are closed over the catheter. Embedding the catheter allows ingrowth of tissue into the cuffs of the catheter with less chance of bacterial colonization. This diminishes the incidence of early pericatheter infections and peritonitis.[16,17] Exteriorization of the catheter is done by making a small incision (1 cm) through the original exit site and using hemostats to locate and bring the catheter and surrounding tunnel to the exit site. The tunnel is incised and the catheter exteriorized.

Catheters buried in this fashion have been successfully exteriorized and used several years after insertion.[18] We recommend embedding the catheter when the catheter will not be used for at least a month, which obviates the need for care of the exit site and flushing the catheter after placement. The procedure allows time for ingrowth of fibrous tissue to the catheter cuffs and allows PD catheter placement to be performed electively rather than urgently (similar to creation of an AVF or AVG).

Complications of Peritoneal Dialysis Catheter Insertion

Bowel perforation is the most feared complication of peritoneal catheter insertion. The incidence is 1% to 1.4% with surgical insertion[2,3] and up to 0.8% with peritoneoscopic or fluoroscopic insertion.[1,3,19] During peritoneoscopic insertions, the diagnosis of bowel perforation is established by direct visualization of bowel mucosa or bowel contents, discharge of fecal material, or emanation of foul-smelling gas through the cannula. In fluoroscopic procedures it is recognized by a characteristic pattern of injected dye outlining the bowel mucosa. Some physicians suggest this complication should be treated surgically,[20] but successful conservative management with bowel rest and intravenous antibiotics is possible if the perforation is recognized early in the procedure (before dilation of the tract or insertion of the catheter).[13,21] To minimize risk for perforation, a needle with a blunt and self-retracting stylet (such as a Veress needle or a Hawkins 18-gauge needle with obturator) can be used instead of a trocar or sharp needle to gain access to the abdominal cavity.[11] Previous abdominal surgery is a relative contraindication to PD catheter placement because of risk of bowel perforation due to intraperitoneal adhesions.[22,23] However, with careful ultrasound examination before needle placement, an area free of adhesions can be identified in almost every patient. If the visceral peritoneal surface moves freely versus the parietal peritoneal surface with each inspiration, then there are no major adhesions in that area of the peritoneum the risk of bowel perforation is low.

Peritoneoscopy or laparoscopy is useful for patients at high risk for adhesions. The scope can be used to identify intraperitoneal adhesions, assess their extent, and locate another site more suitable for catheter placement. With either fluoroscopic or peritoneoscopic techniques, the incidence of bowel perforation is no higher than with surgery, and the early success rate of catheters exceeds 95%.[1,24]

Catheter Repositioning

Outflow failure of a PD catheter occurs whenever the outflow fluid volume is less than the inflow volume. Peritonitis that fails to resolve is the number one reason for removal of PD catheters, but outflow failure is a close second in causes of PD catheter failure. Outflow failure is caused principally by omental attachment to the catheter. If omentum attaches to only a few side holes of the PD catheter, the omentum then acts as a "flap valve" limiting outflow through the PD catheter. As the omentum retracts and moves, the tip of the catheter often migrates to the upper abdomen. However, outflow failure due to adhesions can occur without catheter migration.

A variety of techniques have been used for repositioning PD catheters, including guidewire or stylet insertion, manipulation using a Fogarty catheter, and laparoscopy. Several techniques for catheter repositioning are feasible for nephrologists to perform. Fogarty catheter manipulation is the simplest method and works best with straight Tenckhoff catheters rather than coiled catheters. A Fogarty catheter is advanced past the tip of the catheter, the balloon is inflated, and tugging movements are used to reposition the catheter into the pelvis. Infusion and drainage of dialysate and fluoroscopy are performed to determine patency and position of the PD catheter. Peritoneal catheters can also be repositioned using stiff guidewires or stylets. Fluoroscopic images are helpful in determining the degree of catheter entrapment and the progress of the repositioning procedure.[12] The long-term success rate of repositioned catheters is only 27% to 48%,[25,26] probably because the primary problem is attachment of omentum to the catheter rather than migration of the catheter. Insertion of a new catheter is the most effective therapy in many patients.

Removal of Peritoneal Dialysis Catheters

A Tenckhoff curled or straight PD catheter can be safely removed in an outpatient procedure room; an operating room and general anesthesia are not required.[8] Local anesthetic is infiltrated at the site of the original primary incision, and the incision is incised. Dissection is carried down to the subcutaneous portion of the catheter, which is elevated while the surrounding fibrous sheath is opened. The catheter is clamped with a hemostat, and a nylon suture is placed through the catheter just outside the hemostat. The catheter is cut between the tag and hemostat. Dissection is continued toward the deep cuff by incising the fibrous tunnel adjacent to the catheter (Fig. 97.5). Additional anesthetic is infiltrated around the deep cuff. For catheters that have been in place for less than a month, blunt dissection is usually sufficient to free the deep cuff. Catheters in place for more than a month require sharp dissection, clamping and cautery to free the deep cuff from the rectus muscle. Before removing the deep cuff from the musculature, a purse-string suture is placed into the rectus sheath around the cuff. The deep cuff is then separated from the surrounding tissue, the intraperitoneal portion of the catheter is gently withdrawn from the peritoneal cavity, and the defect in the rectus sheath is closed by tying purse-string suture. The nylon tag is then pulled to expose the remaining subcutaneous portion of catheter segment within the primary incision, and dissection is performed in the direction of the superficial cuff. Once the superficial cuff is free, the external portion of the catheter is cut off at the skin level, and the outer portion of the catheter is removed through the primary incision site. Absorbable suture material in the dermis is used to close the skin incision, reinforced by nylon sutures in the epidermis or sterile adhesive strips if needed. The exit site is not sutured to allow drainage of fluid or pus if it develops.

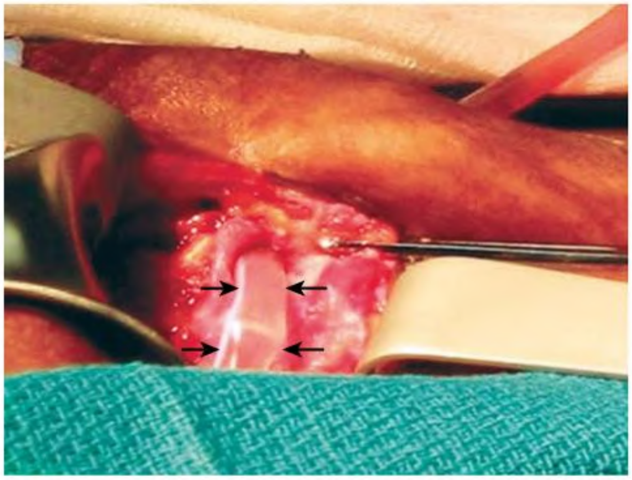

Fig. 97.5 Peritoneal Dialysis Catheter Removal. The catheter (*arrows* show lateral margins of catheter) has been exposed by dissection of the subcutaneous tunnel.

Training and Certification

ASDIN has established training guidelines and criteria for certification of physicians in the insertion of PD catheters by fluoroscopy or peritoneoscopy.[27] In addition to appropriate didactic training, there should be two practice insertions (into models, animals, or human cadavers), observation of two insertions into patients, and then six successful PD catheter insertions into patients with the physician as primary operator.

TUNNELED HEMODIALYSIS CATHETERS

Central venous catheters are used as a temporary HD access, as a bridge to AVF or AVG use, and when all other permanent access sites have been exhausted. Nontunneled catheters are used when a limited number of dialysis sessions is anticipated or there are contraindications to tunneled catheters (systemic infection, risk for bleeding), but they are more appropriate for use in inpatients. Tunneled catheters can be placed in both inpatient and outpatient settings, can be inserted at multiple vein locations, are relatively low in cost, and provide immediate access. However, there are significant disadvantages, including morbidity from infection and thrombosis and the risk of central vein stenosis or occlusion[28,29] (see Chapter 96).

Tunneled Catheter Insertion

The right internal jugular vein is the preferred catheter location compared with the left internal jugular and subclavian vein sites; it provides a straight route to the right atrium, thereby reducing the risk for central vein stenosis. Catheters also may be placed in the femoral veins.

Catheter insertion should be performed in a sterile setting, ideally in an operating room environment with fluoroscopy available or at a minimum in a dedicated procedure room with cardiac monitoring. Before cannulation the vein should be located by ultrasound to detect anatomic variation or venous thrombosis. The patient's neck is then prepared and draped in sterile fashion; under ultrasound guidance, the vein is cannulated with a micropuncture needle (18–22 gauge), and a micropuncture guidewire is inserted and positioned in the superior vena cava. The needle is then removed, and the micropuncture dilator is inserted over the guidewire so it can be replaced with a standard guidewire. The use of the smaller needle rather than the standard 15-gauge needle minimizes trauma to the vein. A small subcutaneous incision is made adjacent to the dilator or guidewire, additional dilation is performed, and the catheter is placed over the guidewire, with care taken to hold the guidewire in place. If a tunneled catheter is to be placed, a catheter exit site is selected inferior to the clavicle and sufficiently lateral to the venotomy to avoid a kink in the catheter. A 1-cm superficial incision is made at this point, and a subcutaneous tract adjacent to the venotomy is infiltrated with lidocaine. A double-lumen catheter, generally 24 or 28 cm in length, is attached to the tunneling device and pulled through the subcutaneous tunnel in a curved path. A guidewire is passed through the dilator and into the inferior vena cava. The venotomy site is then serially dilated over the guidewire. The catheter can then be inserted over the guidewire through the venous port. When a split-tip catheter is used, the guidewire is passed in and out of the two venous ports and through an arterial port or through a hollow intracatheter stiffener. Alternatively, a peel-away sheath is placed over the guidewire and the catheter inserted through the sheath after the removal of the guidewire; however, this method has greater potential for blood loss and air embolism. Fluoroscopy is used to confirm tip placement at the level of the right atrium, with the arterial port facing away from the atrial wall, and to ensure there are no kinks in the catheter (Fig. 97.6). Each port of the catheter is then flushed with saline and locked with the appropriate amount of heparin based on catheter length and priming volume designation, followed by placement of the catheter hub caps.

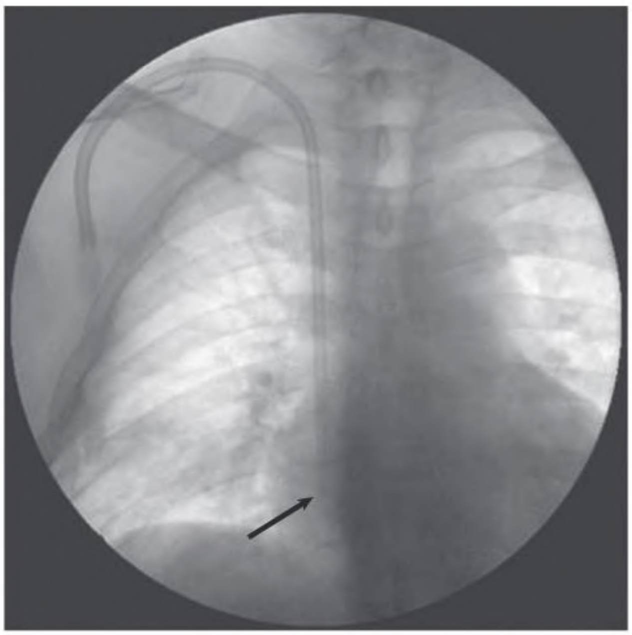

Fig. 97.6 Insertion of a Venous Catheter for Hemodialysis. Chest radiograph confirming that the tip of the catheter *(arrow)* is at the junction of the superior vena cava and the right atrium.

Catheter Dysfunction

Catheter dysfunction is the failure to maintain a blood flow sufficient to perform HD without significantly extending treatment time; this is usually 300 mL/min or greater for conventional dialysis.[30] Causes of immediate dysfunction include a kink in the catheter, incorrect position or orientation (e.g., arterial port against the vessel wall), and errant venous cannulation. These problems should be ascertained and corrected at the time of catheter placement. Catheter thrombosis is the most common cause of late dysfunction. Extrinsic thrombosis is less common than intrinsic thrombosis and is caused by central vein, mural, or right atrial thrombosis. Intrinsic obstruction results from thrombus within the catheter lumen or tip or most commonly from a fibrin sheath. Fibrin sheaths typically develop weeks to months after catheter insertion and result when a sleeve of connective tissue forms at the venotomy site and extends and encases the catheter tip, creating a flap valve. First-line treatment of catheter thrombosis includes forceful flush of the catheter with saline. If flow is not restored, a fibrinolytic agent should be instilled. Tissue plasminogen activator (tPA) is commonly used and appears more effective than urokinase in restoring patency and adequate flow.[31,32] Typically, 2 mg of tPA is instilled per occluded catheter lumen with 0.9% sodium chloride without preservative to fill the internal volume of each lumen and is allowed to dwell 30 minutes or until the next dialysis session. If this fails, the catheter should be exchanged. Strategies to minimize dialysis catheter thrombosis are discussed in Chapter 96.

Catheter Exchange and Fibrin Sheath Removal

Catheter exchange over a guidewire is useful in the setting of catheter thrombosis or bacteremia and allows the preservation of the venotomy, tunnel, and exit sites. The tunnel and exit sites must appear free of infection if the same sites are to be used. Catheter exchange should take place within 72 hours of the initiation of antibiotic therapy.[30] Under sterile conditions, the exit site is anesthetized and the cuff is freed. Once the catheter is pulled back 8 to 10 cm, contrast material is injected through the catheter under fluoroscopy to check for a fibrin sheath (Fig. 97.7). To obliterate a sheath, a guidewire is passed down the venous port of the catheter and into the inferior vena cava. The catheter is then removed, and a balloon catheter is inserted over the guidewire to the sheath location and inflated to disrupt the sheath. A new catheter is then inserted over the guidewire. When the catheter tip is beyond the venotomy site, near the superior vena cava, contrast material can be injected again to check for sheath removal before proceeding with catheter insertion.

Training and Certification

The ASDIN guidelines for HD vascular access procedure certification specify formal didactic training in central venous anatomy, sonographic examination of central veins, fluoroscopy, and catheter design and complications. Practical training for certification includes satisfactory insertion of 25 tunneled long-term catheters (see https://www.asdin.org).

PROCEDURES ON ARTERIOVENOUS FISTULAS AND ARTERIOVENOUS GRAFTS

The most common indications for intervention are inadequate flow during dialysis, thrombosis, and failure of AVFs to mature. Specific interventions include angiography, thrombectomy, angioplasty, and stenting. All of these procedures require a dedicated facility, either

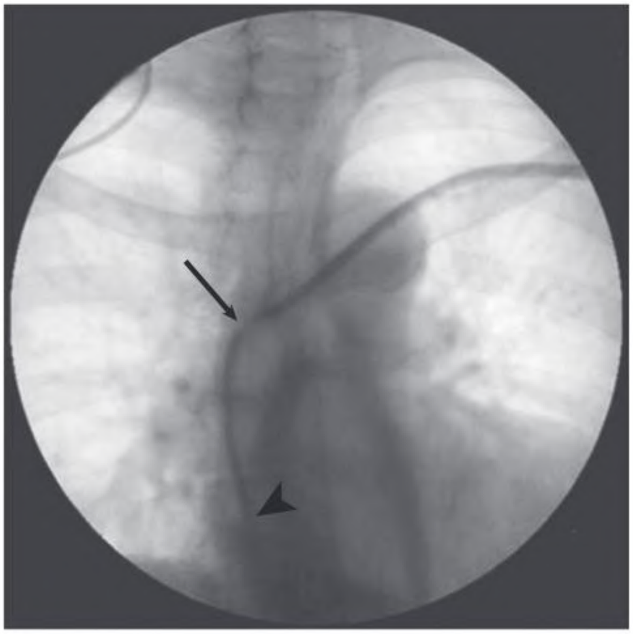

Fig. 97.7 Fibrin Sheath on a Tunneled Venous Catheter. Contrast material has been injected into a tunneled catheter after the tip *(arrow)* has been pulled back into the innominate vein. The contrast material fills a sheath that extends from the catheter tip as far as the *arrowhead.*

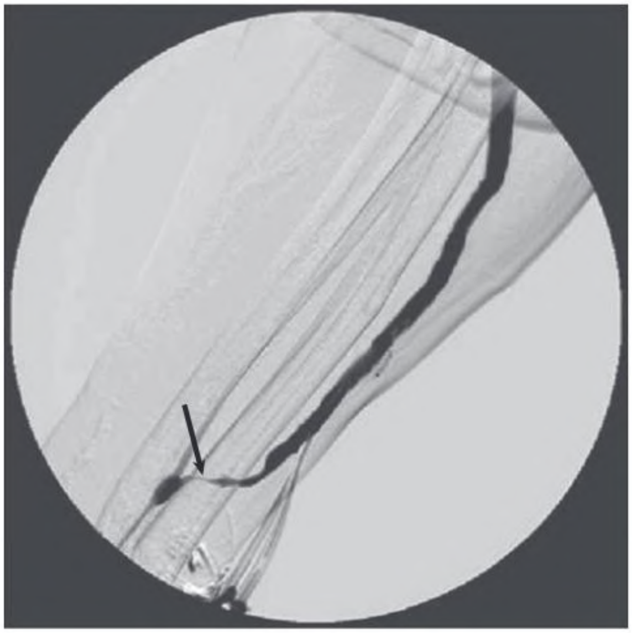

Fig. 97.8 Juxtaanastomotic Stenosis in a Radiocephalic Arteriovenous Fistula. Contrast material was injected at the arterial anastomosis *(bottom left)* and demonstrates a narrowing in the initial portion of the fistula *(arrow).*

inpatient or outpatient, with fluoroscopy, monitoring equipment, and staff to assist with the procedures and deliver conscious sedation. There are many different techniques for AV access procedures and few data to indicate superiority of one method over the other, so the choice is generally one of personal preference and cost. However, the first step always should include careful physical and ultrasound examination of the access. An examination will generally identify the problem and allow detection of access infection, an absolute contraindication to intervention. Appropriate intervention then can be planned. Monitoring and management of vascular access to minimize stenosis, thrombosis, and failure are discussed further in Chapter 96.

Percutaneous Balloon Angioplasty

Stenosis in AVG and AVF is routinely managed by percutaneous balloon angioplasty, which causes minimal discomfort and allows immediate use of the access. Not all stenotic lesions are responsive, however, and some require repeated treatment. In fistulas, the stenosis is most commonly located at the "swing point," including the portion of the native vein mobilized during creation of the AV anastomosis (Fig. 97.8), during vein transposition, or at the cephalic arch; in grafts, the venous anastomosis is the most common site of stenosis.[33-37] Angioplasty is indicated if the stenosis is 50% or more and is associated with clinical or physiologic abnormalities.[30] Treatment of stenosis increases access blood flow and longevity and reduces access thrombosis and access-related hospitalization.[37-39] A relative contraindication to angioplasty is a newly created access (less than 4–6 weeks old).

The access is cannulated with an introducer needle, a sheath is inserted, and initial angiography is performed. This should include views of the access, draining veins (peripheral and central), and arterial anastomosis and is used to confirm the location and degree of stenosis. Unless contraindicated, short-acting sedation and analgesia are given once a lesion has been identified on initial angiography, because angioplasty is painful.

A guidewire is passed through the sheath and across the stenosis. An angioplasty balloon catheter is passed over the guidewire,

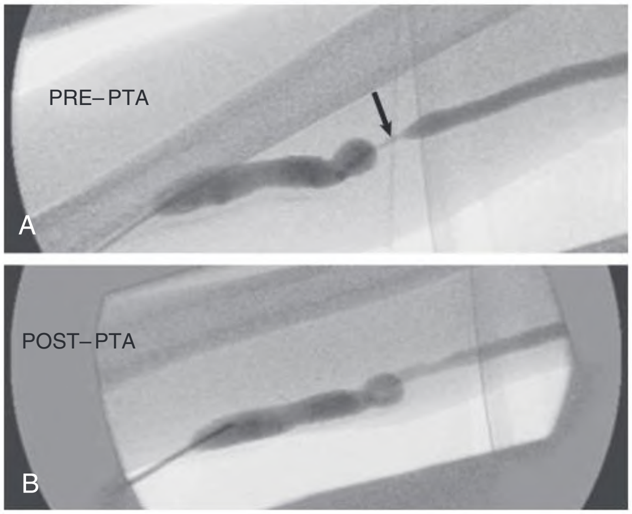

Fig. 97.9 Arteriovenous Graft Stenosis. (A) Stenosis in the outflow vein of an upper arm arteriovenous graft *(arrow).* (B) Angiogram performed immediately after percutaneous angioplasty. *PTA,* Percutaneous transluminal angioplasty.

positioned at the stenotic site, and inflated with a syringe or inflation device (Fig. 97.9). A variety of sheaths, guidewires, balloon sizes, and maximum pressures are available. The guidewire is left in place, and angiography is repeated to identify residual stenosis or any complications. Angioplasty is repeated for residual stenosis or when multiple lesions are present and may require a second cannulation of the access in the opposite direction for inflow stenoses. After removal of all devices, hemostasis at the cannulation site is achieved by manual pressure or suture placement. There is no evidence to support the use of antiplatelet agents or anticoagulation after intervention. According to Kidney Disease Outcomes Quality Initiative guidelines, a successful angioplasty is achieved when there is no more than 30% residual stenosis and physical indicators of stenosis have resolved.[30]

Percutaneous Thrombectomy

A variety of techniques are used for thrombus removal. In thromboaspiration—the least costly method, and as effective and efficient as mechanical and pharmacomechanical thrombolysis—low-dose tPA is instilled into the thrombosed access, the clot is manually macerated, flow returns, and angioplasty is used to dilate access stenoses.[40] Thrombectomy by thromboaspiration combines angiography with balloon angioplasty and thrombectomy by clot aspiration. Absolute contraindications to thromboaspiration include access infection and known right-to-left cardiac shunt; relative contraindications include a large clot burden and long-standing access occlusion.

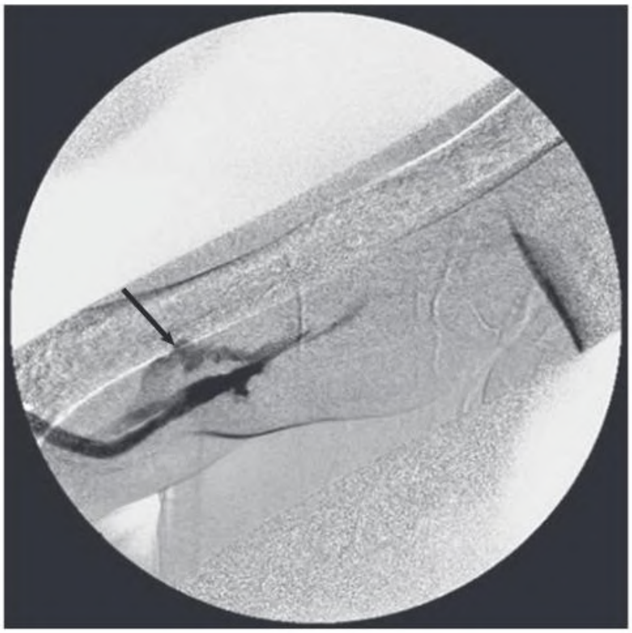

Fig. 97.10 Vein Rupture. Postangioplasty angiogram of an arteriovenous fistula showing extravasation of dye *(arrow),* indicative of a vein rupture. (Courtesy Dr. G. Beathard, Austin, TX.)

The access is cannulated in an antegrade direction, and a guidewire is passed to the level of the central veins. A straight catheter is inserted over the wire to the central veins, and angiography is performed to confirm central venous patency. Anticoagulation and short-acting sedative and analgesic medications are administered in the central circulation. An angiogram is then obtained as the catheter is pulled back to identify the location of stenosis. The guidewire is then inserted beyond the stenotic lesion, followed by an angioplasty balloon catheter. The balloon catheter is insufflated by hand with a syringe or inflation device, and the stenotic lesion is dilated. The access is then cannulated in the retrograde direction, a sheath is inserted, and a Fogarty catheter is passed across the arterial anastomosis, inflated, and pulled back through the entire length of the access while clot fragments are aspirated. On return of flow through the access, angiography is performed to evaluate the inflow and the arterial anastomosis, and angioplasty is repeated if necessary. Hemostasis is achieved by manual pressure or a suture at the cannulation sites.

Stents

Stents are considered in the setting of failed balloon angioplasty (an elastic lesion), when there are few remaining access sites, if the patient is not a surgical candidate for a new access, or when an outflow vein ruptures after balloon angioplasty (Figs. 97.10 and 97.11).[41,42] Results of randomized studies designed to demonstrate stent graft noninferiority to conventional angioplasty for AVG venous anastomotic stenosis show an increased primary patency for stent grafts.[43-45] Further, data suggests improved patency with stent grafts used to treat outflow vein restenosis in AVG, AVF, and central veins.[46-49,50] Finally, a stent may be useful in the setting of an expanding pseudoaneurysm.[51,52]

Despite these encouraging findings, stents should be used judiciously, particularly when surgical options may provide greater enduring patency. Stents should be avoided in situations in which their use will not extend the life of the access.

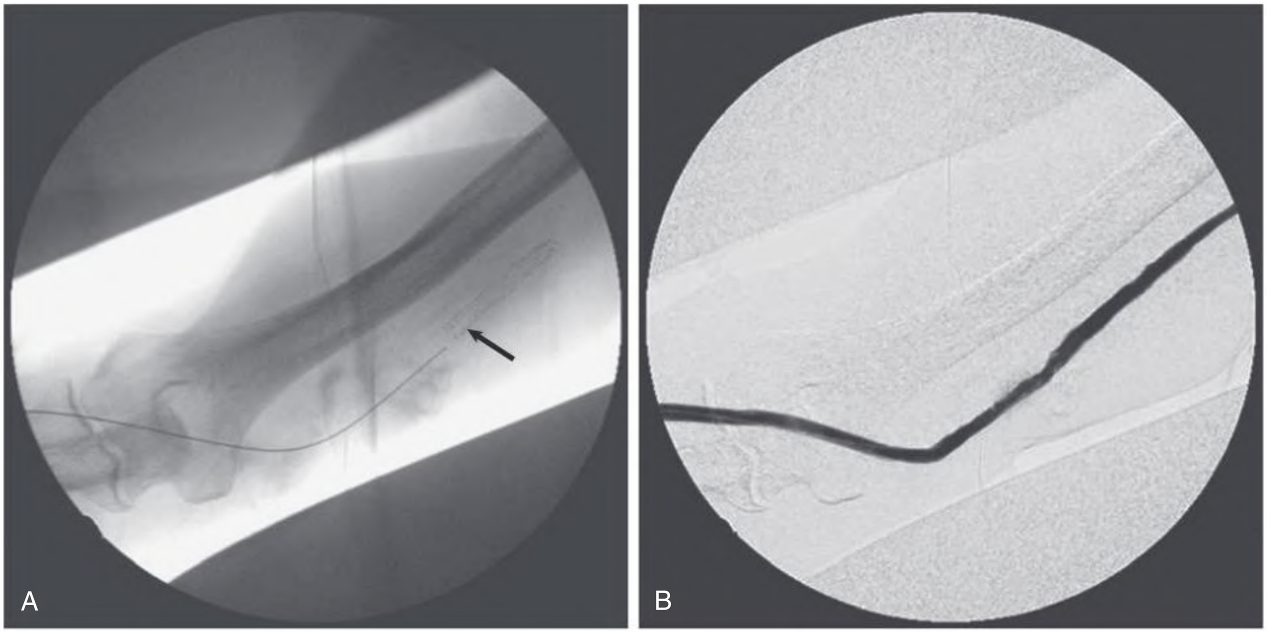

Fig. 97.11 Treatment of Vein Rupture With an Intraluminal Stent. (A) Placement of the stent *(arrow).* (B) An angiogram obtained after stent placement showing that venous outflow has been reestablished. (Courtesy Dr. G. Beathard, Austin, TX.)

Training and Certification

The ASDIN guidelines for HD vascular access procedure certification specify didactic training in venous anatomy, fluoroscopy, procedural equipment, and sedation and analgesia. Requirements for practical training include 125 procedures in both fistulas and grafts of each of the following: angiography, angioplasty, and thrombectomy as primary operator (refer to https://www.asdin.org for more information). In general, several hundred procedures as secondary operator are required before one can become a primary operator.

SELF-ASSESSMENT QUESTIONS

1. tPA is generally effective in treating poor blood flow in tunneled central venous catheters resulting from fibrin sheath formation.
   A. True
   B. False
2. A 57-year-old man with ESKD with a mature left brachiocephalic AVF is referred for evaluation of prolonged bleeding from the AVF on removal of the dialysis needles. The AVF is pulsatile on examination and does not collapse when the left arm is raised above the heart. An angiogram identifies a 75% stenosis at the cephalic arch, and an angioplasty is performed. According to the National Kidney Foundation's Kidney Disease Outcomes Quality Initiative guidelines, what defines a successful percutaneous angioplasty of an arteriovenous access?
   A. Less than 10% residual stenosis at the angioplasty site
   B. Ability to receive eight uncomplicated, consecutive dialysis sessions after the procedure
   C. Less than 30% residual stenosis and resolution of physical indicators of stenosis
   D. A and B
   E. B and C
3. Embedding (or burying) a newly placed PD catheter under the skin:
   A. allows ingrowth of tissue into the cuffs of the catheter without an opportunity for bacterial colonization.
   B. diminishes the incidence of early pericatheter infections.
   C. results in better catheter function in the future.
   D. A and B.
   E. is of no benefit other than cosmetic.
4. Which of the following statements about ultrasound of the renal and urinary tract is *false?*
   A. For examination of the kidneys, a renal ultrasound is always indicated in the workup of chronic kidney disease.
   B. A renal ultrasound is not indicated in every patient with acute kidney injury.
   C. Ultrasound is not useful in the diagnosis of bladder outlet obstruction.
   D. The ureters are usually visible only when dilated.
5. Insertion of PD catheters is commonly performed using:
   A. laparoscopy.
   B. surgical dissection.
   C. peritoneoscopy.
   D. the guidewire (Seldinger) technique.
   E. all the above.

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